

OpenEXR Attributes Base Specification for Camera Acquisition Metadata

Version: 0.5 DRAFT 2

Date: 2026-07-19

Status: FOR REVIEW

Author: Sony Corporation

1. General

1.1 Scope

This specification defines the format for describing metadata as OpenEXR Header Attributes, where the metadata is extracted from MXF clips captured by video cameras or from signals output by cameras and other peripheral devices, hereinafter referred to as Acquisition Metadata. It also addresses the format for describing such metadata in USD (Universal Scene Description) format.

The purpose of this Base Specification is to provide common metadata rules that can be used across multiple camera manufacturers, acquisition formats, and production pipelines.

Editor's Note:

This document is STILL in the draft stage. Various comments from reviewers are welcome in order to develop this into a more practical specification.

1.2 Source Metadata

This specification mainly targets the following as information to be stored in the OpenEXR Header Attribute:

- Structural metadata contained in the Header Metadata of the MXF file
- Metadata contained as a System Item or Data Item in the Essence Container of the MXF file
- Metadata recorded by the camera as a file separate from the MXF file
- Metadata contained in signals obtained from the camera via SDI or an IP network
- Metadata acquired by a device other than the camera

In this document of this specification, specific methods for referencing source information are defined only for MXF structural metadata. This document does not address methods for referencing other source information; rather, assuming that such information has been obtained by some means, it defines the format for storing that information as OpenEXR attributes. Methods for referencing such source information may be defined in separate documents constituting this specification.

1.3 OpenEXR Header Attributes

1.3.1 OpenEXR Specification

The OpenEXR header attributes defined in this specification shall conform to version 3.x of the OpenEXR specification. This can be obtained from the following:

- OpenEXR documents (<https://openexr.com/en/latest/index.html>)

Equivalent specifications are also defined in the following:

- SMPTE ST 2065-4:2023 ACES Image Container File Layout

1.3.2 Overview of OpenEXR Header Attributes

OpenEXR file starts with a 4-byte magic number, and consists of a 4-byte version field, header, line offset table, and scan line blocks. The header can contain multiple metadata as header attributes. The metadata described in this document is also recorded in this area. The header area also contains header attributes predefined by OpenEXR other than those described in this document, but the metadata described in this document is written before those.

An OpenEXR Header Attribute consists of the following elements:

- Attribute name,
- Attribute type name,
- Attribute size, and
- Attribute value.

1.4 Data Types

In this specification, types compliant with the OpenEXR specification SHALL be used for OpenEXR attributes.

Note that it is safer to avoid using the bytes type added in OpenEXR v3.4, as some applications do not currently support it.

1.4.1 Timestamp

When an OpenEXR attribute defined in this specification represents a date and time, unless otherwise specified, the value SHALL be represented as a string compliant with ISO 8601 using the string type. Such strings include, for example, the following:

- `2026-07-18T12:38:55-07:00` (local time representation with a time zone)
- `2026-07-18T19:38:55Z` (UTC representation)

Editor's Note:

The OpenEXR standard attribute `capDate` represents local time without a time zone, and a separate attribute, `utcOffset`, is used to indicate the time zone. In this specification, the method described above was considered appropriate because the time zone can be clearly identified from the timestamp alone.

Editor's Note:

The `T` character used to separate the date and time in ISO 8601 form is somewhat visually difficult to read. Should RFC 3339 compliance be required?

1.4.2 Universal Label, UMID and UUID

Universal Labels, UMIDs, UUIDs, and similar identifiers are 16-byte values. However, because OpenEXR does not define an appropriate type for representing them, they shall be stored as 32-character hexadecimal strings such as:

```
060A2B340101010501010D4313000000A8202CAA398605C508004602170707F2
```

The characters A through F shall be written in uppercase, and readers shall accept both uppercase and lowercase characters.

Editor's Note:

The sequence of bytes above is easier to parse, but should URN syntax be followed for human readability?

1.4.3 Rectangle

In MXF, a rectangle indicating a picture area shall be stored as a box2i type in the following format. Here, XOffset and YOffset are the numbers of offset pixels, with the upper-left corner as the origin and the rightward and downward directions defined as positive, respectively. Width and Height are the numbers of pixels of the rectangle in the horizontal and vertical directions, respectively.

- `xMin` = `XOffset`
- `xMax` = `XOffset` + `Width` - 1
- `yMin` = `YOffset`
- `yMax` = `YOffset` + `Height` - 1

2. Convention Policy

This section describes the policy for defining OpenEXR attributes.

2.1 Name Space

The OpenEXR attribute names defined in this specification SHALL be formed by concatenating the basename immediately after the namespace identifier, as follows:

`namespace_identifier` `basename`

The following sections define the namespaces and their formats.

2.1.1 Delimiter

A namespace may have a hierarchical structure. In this specification, in the character string constituting an OpenEXR attribute name, a colon (`:`) SHALL be used as the delimiter character for separating namespace hierarchy levels.

Editor's Note:

Depending on the environment, a colon may not be appropriate. Possible alternatives include a period, although this may also be undesirable, a slash, or an underscore. Further consideration is required.

2.1.2 Character Set

Except for delimiter characters, namespace identifiers SHALL be composed of lowercase ASCII characters and shall not contain whitespace characters, unless there is a specific reason to do otherwise.

If it is necessary to separate words within a hierarchy level in a namespace identifier, the use of an underscore (`_`) is recommended.

2.1.3 Top-Level Namespace

All OpenEXR attribute names to be defined in this specification SHALL begin with `acq:` .

2.1.4 Sub-Namespace

A sub-namespace indicating the classification of Acquisition Metadata MAY be written after `acq:` as needed. A sub-namespace MAY further contain sub-namespaces under it.

This document defines the following as second-level sub-namespaces.

- MXF sub-namespace (`mxf:`)
- Lens sub-namespace (`lens:`)
- Camera sub-namespace (`camera:`)
- Imager sub-namespace (`imager:`)
- Color sub-namespace (`color:`)
- Slate sub-namespace (`slate:`)

For the meanings of the respective namespaces, refer to Section 3.1.

2.1.5 Addition of a New Namespace

Extension specifications based on this specification MAY define their own dedicated sub-namespaces.

For example, the namespace identifier `acq:abc:` MAY be defined and used for attributes that store information conforming to a standard named ABC. Similarly, the namespace identifier `acq:xyz:` MAY be defined and used for attributes that store information specific to a vendor named XYZ.

2.1.6 Namespace Schema Identification

Each namespace MAY have a Schema Identifier that describes the namespace.

The Schema Identifier SHALL be represented by an attribute name formed by appending the sub-namespace identifier `schema:` and a base name immediately after the target namespace identifier.

In the top-level namespace defined by this specification, attributes describing this specification shall be specified as shown in Section 3.2.1

In addition, in namespaces extended in accordance with Section 2.1.5, attributes describing the subject matter handled by the respective namespace shall be specified as shown in Section 3.2.2.

2.2 Preference for Unified Attributes

As described in Section 2.1.5, this specification allows namespaces to be extended and attributes to be defined for vendor-specific metadata.

However, metadata representing common concepts should be normalized and stored as vendor-neutral attributes defined in this document. In such cases, the vendor-specific attribute may also be stored together with it.

If there is a concern that information may be lost as a result of normalization into vendor-neutral attributes, the vendor-specific information should be referenced as necessary.

Furthermore, if metadata can be represented using OpenEXR standard attributes, it is recommended that those attributes be used. In such cases, vendor-neutral attributes defined in this specification and vendor-specific attributes MAY also be stored together with them.

3 Metadata Mapping

3.1 Overview

The vendor-neutral attributes defined in this specification are shown in the table below.

Namespace Identifier	Description for the source information
<code>acq:schema:</code>	Schema Identification for the top-level namespace, that is, this specification.
<code>acq:translator:</code>	Information retained or generated by applications that convert MXF files to OpenEXR.
<code>acq:mxf:</code>	Vendor-neutral Acquisition Metadata extracted from MXF structural metadata. The scope is restricted only to properties compliant with the MXF specification (SMPTE ST 377-1) and related public specifications. Vendor-specific extended sub-descriptors SHALL NOT be included.
<code>acq:lens:</code>	Vendor-neutral normalized Acquisition Metadata that is related to the camera device used for lens.
<code>acq:camera:</code>	Vendor-neutral normalized Acquisition Metadata that is related to the camera device used for capture.
<code>acq:imager:</code>	Vendor-neutral normalized Acquisition Metadata that is related to the imager device used for capture.
<code>acq:color:</code>	Vendor-neutral normalized Acquisition Metadata that is related to the color processing used for capture.
<code>acq:slate:</code>	Vendor-neutral slate metadata.

Additional Sub-namespace Identifiers may be appended after the Namespace Identifier described above, as needed.

Editor's Note:

`imager:` and `color:` are subdivisions of `camera:` ; however, there is a concern that such subdivision may make the hierarchy deeper and may also make classification more difficult.

3.2 Schema Identifier Attributes

3.2.1 Top-Level Schema Identifier Attributes

The following attributes, which describes this specification itself, SHALL be specified as the top-level Schema Identifier.

Attribute Name	Type	Requirement	Value
<code>acq:schema:standard</code>	<code>string</code>	REQUIRED	"Open Acquisition Metadata"
<code>acq:schema:version</code>	<code>string</code>	REQUIRED	"0.50"

3.2.2 Vendor Extension Schema Identifier Attributes

When a namespace is extended to store vendor-specific metadata or metadata conforming to a particular specification, the following attribute, which describes that namespace, shall be specified.

Here, *extended_sub_namespace* denotes the identifier of the extended namespace.

Attribute Name	Type	Requirement	Value
<code>acq: <i>extended_sub_namespace</i> :schema:standard</code>	<code>string</code>	REQUIRED	Name of the standard, or vendor

Attribute Name	Type	Requirement	Value
<code>acq:extended_sub_namespace</code> <code>schema:version</code>	<code>string</code>	REQUIRED	Version of the standard.

3.3 Translator Attributes

The following OpenEXR attributes are defined to describe information retained or generated by the converter that converts the source MXF file to OpenEXR.

Attribute Name	Type	Requirement	Description
<code>acq:translator:clipName</code>	<code>string</code>	OPTIONAL	File name of the source MXF file
<code>acq:translator:appName</code>	<code>string</code>	OPTIONAL	Name of the application that created this OpenEXR file
<code>acq:translator:appVersion</code>	<code>string</code>	OPTIONAL	Version of the application that created this OpenEXR file
<code>acq:translator:creationDate</code>	<code>string</code> (see 1.4.1)	OPTIONAL	Date and time in Timestamp format when this OpenEXR file was created.

3.4 MXF Attributes

`mxf:` shall be used as the sub-namespace identifier to store Acquisition Metadata extracted from MXF header metadata that is specified in the MXF specification (SMPTE ST 377-1). That is, the OpenEXR attribute names shall begin with `acq:mxf:`.

3.4.1 MXF General Properties

The following OpenEXR attributes are defined to describe general information obtained from the source MXF file.

Attribute Name	Type	Requirement	Description
<code>acq:mxf:creationDate</code>	<code>string</code> (see 1.4.1)	OPTIONAL	The creation date and time of the source MXF file. This value SHALL be obtained from the Package Creation Date item of the Source Package in the Header Metadata of the source MXF file, not the file system timestamp.
<code>acq:mxf:umid</code>	<code>string</code> (see 1.4.2)	OPTIONAL	The UMID of the source MXF file. This value SHALL be obtained from the Package Creation UID item of the Material Package.

3.4.2 MXF Timeline Properties

The following OpenEXR attributes are defined to describe information related to the timeline of the source MXF file.

Attribute Name	Type	Requirement	Description
<code>acq:mxf:timeline:startTimeCode</code>	<code>unsigned long</code>	OPTIONAL	The recording start timecode of the source MXF file. This value SHALL be obtained from the Start Timecode item of the Timecode Component in the Material Package.
<code>acq:mxf:timeline:duration</code>	<code>string</code>	OPTIONAL	The duration of the source MXF file as the number of Edit Units. This value SHALL be obtained from the Duration item of the Sequence of the Timeline Track for Timecode (Track ID = 1) in the Material Package in the Header Metadata of the source MXF file.
<code>acq:mxf:timeline:dropFrame</code>	<code>unsigned char</code>	OPTIONAL	The dropframe flag of the source MXF file. This value SHALL be obtained from the Drop Frame item of the Sequence of the Timeline Track for Timecode (Track ID = 1) of the Timecode Component in the Material Package.
<code>acq:mxf:timeline:editRate</code>	<code>rational</code>	OPTIONAL	The Edit Rate of the source MXF file. This value shall be obtained from the Edit Rate item of the Timeline Track for Timecode (Track ID = 1) of the Material Package.

Among the properties shown in the table above, those listed in the table below are mutually convertible with standard OpenEXR attributes. It is preferable to describe these as standard OpenEXR attributes; however, since the standard attributes may be determined by factors other than this specification, the attributes in the table above may also be recorded together.

MXF Timeline Attribute	OpenEXR Standard Attribute	Conversion
<code>acq:mxf:timeline:editRate</code>	<code>framesPerSecond</code>	

3.4.3 MXF Video Properties

The following OpenEXR attributes are defined to describe information related to the video of the source MXF file.

Attribute Name	Type	Requirement	Description
<code>acq:mxf:video:codec</code>	string	OPTIONAL	A human-readable string indicating the video codec of the source MXF file. This value SHALL be derived from the Essence Container item and the Picture Essence Coding item of the Picture Essence Descriptor.
<code>acq:mxf:video:resolution</code>	v2i	OPTIONAL	The display width and height of the video in the source MXF file. This value SHALL be stored as a v2i type, in order, using the values obtained from the DisplayWidth item and the DisplayHeight item of the Picture Essence Descriptor.
<code>acq:mxf:video:displayRectangle</code>	box2i	OPTIONAL	The position and size of the Display Rectangle of the source MXF file. This value SHALL be stored as a <code>box2i</code> type in accordance with 1.4.3, using the information from DisplayXOffset, DisplayYOffset, DisplayWidth, and DisplayHeight items of the Picture Essence Descriptor.
<code>acq:mxf:video:sampledRectangle</code>	box2i	OPTIONAL	The position and size of the Sampled Rectangle of the source MXF file. This value SHALL be stored as a <code>box2i</code> type in accordance with 1.4.3, using the information from SampledXOffset, SampledYOffset, SampledWidth, and SampledHeight items of the Picture Essence Descriptor.
<code>acq:mxf:video:storedRectangle</code>	box2i	OPTIONAL	The position and size of the Stored Rectangle of the source MXF file. This value SHALL be stored as a <code>box2i</code> type in accordance with 1.4.3, using the information from StoredXOffset, StoredYOffset, StoredWidth, and StoredHeight items of the Picture Essence Descriptor.
<code>acq:mxf:video:aspectRatio</code>	rational	OPTIONAL	The aspect ratio of the video in the source MXF file. This value shall be obtained from the Aspect Ratio item of the Picture Essence Descriptor.
<code>acq:mxf:video:sampleRate</code>	rational	OPTIONAL	The sample rate of the video in the source MXF file. This value shall be obtained from the Sample Rate item of the Picture Essence Descriptor.
<code>acq:mxf:video:frameLayout</code>	unsigned char	OPTIONAL	The frame layout of the video in the source MXF file. This value shall be obtained from the Frame Layout item of the Picture Essence Descriptor.
<code>acq:mxf:video:transferCharacteristic</code>	string	OPTIONAL	The transfer characteristic of the video in the source MXF file. This value SHALL be the string obtained by converting the Universal Label obtained from the Transfer Characteristic item of the Picture Essence Descriptor in accordance with 1.4.2.
<code>acq:mxf:video:codingEquations</code>	string	OPTIONAL	The coding equations of the video in the source MXF file. This value SHALL be the string obtained by converting the Universal Label obtained from the Coding Equations item of the Picture Essence Descriptor in accordance with 1.4.2.
<code>acq:mxf:video:colorPrimaries</code>	string	OPTIONAL	The color primaries of the video in the source MXF file. This value SHALL be the string obtained by converting the Universal Label obtained from the Color Primaries item of the Picture Essence Descriptor in accordance with 1.4.2.

Among the properties shown in the table above, those listed in the table below are mutually convertible with standard OpenEXR attributes. It is preferable to describe these as standard OpenEXR attributes; however, since the standard attributes may be determined by factors other than this specification, the attributes in the table above may also be recorded together.

MXF Timeline Attribute	OpenEXR Standard Attribute	Conversion
<code>acq:mxf:video:sampleRate</code>	<code>framesPerSecond</code>	Fallback only when <code>mxf:timeline:editRate</code> is absent

3.4.4 MXF Audio Properties

The following OpenEXR attributes are defined to describe information related to the audio of the source MXF file.

Attribute Name	Type	Requirement	Description
<code>acq:mxf:numOfAudioCh</code>	unsigned int	OPTIONAL	The number of audio tracks in the source MXF file. This value SHALL be derived from the number of tracks that are Sound Essence Tracks among the track array described as the Tracks item of the Material Package.
<code>acq:mxf:audio:ch N :c odec</code>	string	OPTIONAL	A human-readable string indicating the audio codec of the source MXF file. This value SHALL be derived from the Essence Container item and the Sound Essence Coding item of the Sound Essence Descriptor in the Header Metadata of the source MXF file.
<code>acq:mxf:audio:ch N :s amplingRate</code>	rational	OPTIONAL	The sampling rate of the audio in the source MXF file. This value shall be obtained from the Audio Sampling Rate item of the Sound Essence Descriptor.
<code>acq:mxf:audio:ch N :e ssenceCoding</code>	string	OPTIONAL	The sound essence coding of the source MXF file. This value SHALL be the string obtained by converting the Universal Label of the Sound Essence Coding item of the Sound Essence Descriptor in accordance with 1.4.2.
<code>acq:mxf:audio:ch N :q uantizationBits</code>	string	OPTIONAL	The sound quantization bits of the source MXF file. This value SHALL be obtained from the Quantization Bits item of the Sound Essence Descriptor.

In the table above, `ch N :` is a sub-namespace identifier corresponding to the audio channel number, which shall start from `ch1:`.

3.5 Vendor-Neutral Acquisition Attributes

This section defines vendor-independent attributes. The values of these attributes are assumed to be derived from some standard specification or vendor-specific metadata; however, the specific sources are not specified in this document.

3.5.1 Lens Attributes

Vendor-Neutral Lens Attributes SHALL be defined as follows. These attribute names SHALL use the `acq:lens:` namespace identifier.

Attribute Name	Type	Requirement	Unit	Description
<code>acq:lens:make</code>	string	RECOMMENDED	none	Lens manufacturer
<code>acq:lens:model</code>	string	RECOMMENDED	none	Lens model name
<code>acq:lens:serialNumber</code>	string	RECOMMENDED	none	Serial number of the individual lens unit
<code>acq:lens:firmwareVersion</code>	string	OPTIONAL	none	Lens firmware version
<code>acq:lens:uuid</code>	string	OPTIONAL	none	UUID that identifies the individual lens unit
<code>acq:lens:label</code>	string	OPTIONAL	none	Human-readable string that identifies the individual lens unit
<code>acq:lens:nominalFocalLength</code>	float	OPTIONAL	mm	Nominal focal length
<code>acq:lens:pinholeFocalLength</code>	float	OPTIONAL	mm	Pinhole focal length

Attribute Name	Type	Requirement	Unit	Description
<code>acq:lens:effectiveFocalLength</code>	float	RECOMMENDED	mm	Effective physical focal length
<code>acq:lens:entrancePupilOffset</code>	float	OPTIONAL	mm	Entrance pupil offset
<code>acq:lens:irisFNumber</code>	float	OPTIONAL	F-number	Iris F-number
<code>acq:lens:irisTStop</code>	float	OPTIONAL	T-stop	Iris T-stop
<code>acq:lens:focusDistance</code>	float	RECOMMENDED	mm	Focus distance
<code>acq:lens:opticalExtenderMagnification</code>	int	OPTIONAL	percent	Optical extender magnification
<code>acq:lens:opticalInversion</code>	int	OPTIONAL	code	Optical inversion state
<code>acq:lens:opticalImageStabilization</code>	int	OPTIONAL	1/0	Optical image stabilization flag
<code>acq:lens:macroSetting</code>	int	OPTIONAL	1/0	Macro setting flag

Among the properties shown in the table above, those listed in the table below are mutually convertible with standard OpenEXR attributes. It is preferable to describe these as standard OpenEXR attributes; however, since the standard attributes may be determined by factors other than this specification, the attributes in the table above may also be recorded together.

Vendor-Neutral Lens Attribute	OpenEXR Standard Attribute	Conversion
<code>acq:lens:make</code>	<code>lensMake</code>	no conversion
<code>acq:lens:model</code>	<code>lensModel</code>	no conversion
<code>acq:lens:serialNumber</code>	<code>lensSerialNumber</code>	no conversion
<code>acq:lens:firmwareVersion</code>	<code>lensFirmwareVersion</code>	no conversion
<code>acq:lens:nominalFocalLength</code>	<code>nominalFocalLength</code>	no conversion
<code>acq:lens:pinholeFocalLength</code>	<code>pinholeFocalLength</code>	no conversion
<code>acq:lens:effectiveFocalLength</code>	<code>effectiveFocalLength</code>	no conversion
<code>acq:lens:entrancePupilOffset</code>	<code>entrancePupilOffset</code>	no conversion
<code>acq:lens:irisFNumber</code>	<code>aperture</code>	no conversion
<code>acq:lens:irisTStop</code>	<code>tStop</code>	no conversion
<code>acq:lens:focusDistance</code>	<code>focus</code>	mm to meters

3.5.2 Camera Attributes

These attribute names SHALL generally use the `acq:camera:` namespace identifier; attributes related to the image sensor shall use `acq:imager:`, and attributes related to color processing SHALL use `acq:color:`, respectively.

Attribute Name	Type	Requirement	Unit	Description
<code>acq:camera:make</code>	string	RECOMMENDED	none	Camera manufacturer

Attribute Name	Type	Requirement	Unit	Description
acq:camera:model	string	RECOMMENDED	none	Camera model name
acq:camera:serialNumber	string	RECOMMENDED	none	Serial number of the individual camera unit
acq:camera:firmwareVersion	string	OPTIONAL	none	Camera firmware version
acq:camera:uuid	string	OPTIONAL	none	UUID that identifies the individual camera unit
acq:camera:label	string	OPTIONAL	none	Human-readable string that identifies the individual camera unit
acq:imager:overallDimension	v2f	OPTIONAL	mm	Physical width and height of image sensor
acq:imager:effectiveDimension	v2f	RECOMMENDED	mm	Physical width and height of the effective area within the image sensor
acq:imager:effectiveRectangle	box2i	OPTIONAL	pixels	Acquisition rectangle
acq:imager:pixelPitch	float	OPTIONAL	micrometers	Photosite pitch
acq:camera:ndFilterIndex	int	OPTIONAL	index	Internal ND filter index
acq:camera:ndFilterDensity	float	OPTIONAL	optical density	ND filter optical density
acq:camera:isoSensitivity	float	OPTIONAL	ISO	Actual ISO sensitivity
acq:camera:exposureIndex	float	RECOMMENDED	EI	Exposure index
acq:camera:exposureTime	rational1	RECOMMENDED	rational seconds	Exposure duration
acq:camera:shutterAngle	float	RECOMMENDED	degree	Shutter angle
acq:camera:captureFrameRate	rational1	RECOMMENDED	rational fps	Capture frame rate
acq:color:colorTemperature	int	RECOMMENDED	kelvin	Correlated color temperature
acq:color:tint	float	RECOMMENDED	numeric	Green/Magenta correction
acq:color:captureGammaEquation	string	OPTIONAL	none	Capture gamma or transfer function
acq:color:captureColorPrimaries	string	OPTIONAL	none	Capture color primaries
acq:color:codingEquations	string	OPTIONAL	none	Coding equations
acq:color:luminanceCodeRange	int	OPTIONAL	none	Luminance code range
acq:color:ascCdlFilename	string	OPTIONAL	none	Filename of ASC CDL
acq:color:lookLutFilename	string	OPTIONAL	none	Filename of Look LUT

All of the properties shown in the table above are equivalent to the standard OpenEXR attributes shown in the table below. It is preferable to describe them as standard OpenEXR attributes; however, since the standard attributes may be determined by factors other than this specification, the attributes in the table above may also be recorded together.

Vendor-Neutral Camera Attribute	OpenEXR Standard Attribute	Conversion
acq:camera:make	cameraMake	no conversion
acq:camera:model	cameraModel	no conversion
acq:camera:serialNumber	cameraSerialNumber	no conversion
acq:camera:firmwareVersion	cameraFirmwareVersion	no conversion
acq:camera:uuid	cameraUuid	no conversion
acq:camera:label	cameraLabel	no conversion
acq:imager:overallDimension	sensorOverallDimensions	no conversion
acq:imager:effectiveRectangle	sensorAcquisitionRectangle	no conversion
acq:imager:pixelPitch	sensorPhotositePitch	no conversion
acq:camera:isoSensitivity	isoSpeed	no conversion
acq:camera:exposureTime	expTime	Parse rational seconds to float
acq:camera:shutterAngle	shutterAngle	no conversion
acq:camera:captureFrameRate	captureRate	no conversion
acq:color:colorTemperature	cameraCCTSetting	int to float
acq:color:tint	cameraTintSetting	T.B.D.

3.6 Slate Attributes

Slate attributes shall be defined as follows. These attribute names shall use the `acq:slate:` namespace identifier.

Attribute Key	Type	Requirement	Description
acq:slate:clipName	string	OPTIONAL	Clip name
acq:slate:clipNumber	int	OPTIONAL	Clip number
acq:slate:cameraIndex	string	OPTIONAL	Camera index
acq:slate:reelName	string	OPTIONAL	Reel name
acq:slate:reelNumber	int	OPTIONAL	Reel number
acq:slate:scene	string	OPTIONAL	Scene
acq:slate:take	string	OPTIONAL	Take
acq:slate:circleTake	int	OPTIONAL	Circle take flag
acq:slate:production	string	OPTIONAL	Production name
acq:slate:productionCompany	string	OPTIONAL	Production company
acq:slate:director	string	OPTIONAL	Director

Attribute Key	Type	Requirement	Description
<code>acq:slate:cinematographer</code>	<code>string</code>	OPTIONAL	Cinematographer
<code>acq:slate:location</code>	<code>string</code>	OPTIONAL	Location
<code>acq:slate:episode</code>	<code>string</code>	OPTIONAL	Episode
<code>acq:slate:season</code>	<code>string</code>	OPTIONAL	Season
<code>acq:slate:shootingDay</code>	<code>int</code>	OPTIONAL	Shooting day

4. USD Pipeline Mapping

This section describes the method for deriving USD Metadata and Attributes.

4.1 USD Stage Metadata

4.1.1 `timeCodesPerSecond`

They shall be applied in the following order of priority:

- `mxf:timeline:editRate`
- `mxf:video:sampleRate` , only when `mxf:video:editRate` is absent
- OpenEXR Standard Attribute `framesPerSecond` , only when no authoritative MXF timing source exists

4.1.2 `framesPerSecond`

They shall be applied in the following order of priority:

- `acq:camera:captureFrameRate`
- `mxf:timeline:editRate`
- `mxf:video:sampleRate`
- OpenEXR Standard Attribute `framesPerSecond`

If no capture frame rate is available, `framesPerSecond` SHOULD be set to the same value as `timeCodesPerSecond` .

4.1.3 Timecode

Timecode-related UsdStage metadata shall be derived in accordance with the table below

UsdStage Metadata	OpenEXR Attribute
<code>startTimeCode</code>	<code>mxf:timeline:startTimeCode</code>
<code>endTimeCode</code>	<code>mxf:timeline:startTimeCode</code> + <code>mxf:timeline:duration</code> - 1 frame
May be used for timecode interpretation	<code>mxf:timeline:dropFrame</code>

The transcoder SHALL convert SMPTE timecode strings to numeric USD `timeCode` values using the selected timeline rate.

4.2 UsdGeom.Camera Built-in Attributes

UsdGeom.Camera Built-in Attributes shall be derived in accordance with the table below

UsdGeom.Camera Attribute	OpenEXR Attribute
<code>focalLength</code>	<code>acq:lens:effectiveFocalLength</code> , then <code>acq:lens:nominalFocalLength</code>
<code>fStop</code>	<code>acq:lens:irisFNumber</code>

UsdGeom.Camera Attribute	OpenEXR Attribute
<code>focusDistance</code>	<code>acq:lens:focusDistance</code>
<code>horizontalAperture</code>	<code>acq:sensor:width</code>
<code>verticalAperture</code>	<code>acq:sensor:height</code>
<code>horizontalAperture / verticalAperture</code>	<code>acq:imager:effectiveDimension</code>
<code>shutter:open / shutter:close</code>	<code>acq:camera:exposureTime</code>
<code>exposure:iso</code>	<code>isoSpeed</code>
<code>exposure:time</code>	<code>expTime</code>
<code>xformOp:transform</code>	<code>worldToCamera</code>

4.2.1 Shutter Mapping

When `acq:camera:exposureTime` is selected, it SHALL be parsed as rational seconds.

```
shutterDurationFrames = exposureTimeSeconds * framesPerSecond
shutter:open = -shutterDurationFrames / 2
shutter:close = shutterDurationFrames / 2
```

4.3 Consumed Attributes

Attributes selected as sources for USD Stage Metadata or `UsdGeom.Camera` built-in attributes SHALL be treated as consumed attributes.

Unselected candidate attributes MAY be preserved in `customData` .

If preservation of consumed source values is required, implementations MAY store them under:

```
sourceAttributes:<originalAttributeName>
```

Examples:

```
sourceAttributes:mxf:timeline:editRate = "24000/1001"
sourceAttributes:acq:camera:exposureTime = "1001/60000"
sourceAttributes:acq:lens:effectiveFocalLength = 50.0
```

